

The Art of Touch-n-Curl: Designing Curvilinear Sculptures via Interlocking 3D Printed Zippers

Fanke Qi
Zhejiang University
Hangzhou, China
fankeqi@u.nus.edu

Fanqi Zhou
Zhejiang University
Hangzhou, China
fanqizhou@zju.edu.cn

Ruoyu Zhai
College of Science and Technology
Ningbo University
Ningbo, China
3193655075@qq.com

Deying Pan
Zhejiang University
Hangzhou, China
deyingp2@zju.edu.cn

Yongbo Ni
Zhejiang University
Hangzhou, China
sparksnyb@gmail.com

Tianshu Dong
Zhejiang University
Hangzhou, China
dongts@zju.edu.cn

Yitao Fan
Zhejiang University
Hangzhou, China
ytfan@zju.edu.cn

Guanyun Wang*
Zhejiang University
Hangzhou, China
guanyun@zju.edu.cn

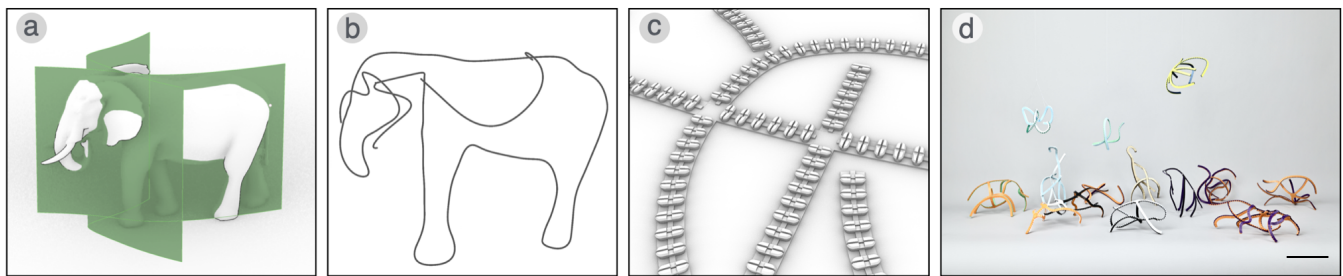


Figure 1: A preview of Curve Aesthetics (a-d): (a) mesh as input (cutting surfaces marked in green), (b) space curves as output, (c) ribbons and arrayed zipper teeth, and (d) animal sculptures. (Scale Bar: 12 cm.)

Abstract

This demo presents a design suite of curvilinear animal sculptures using Touch-n-Curl, a design and construction pipeline that encodes space curves with a pair of 3D printed flat zippers manually interlocking together in space into corresponding 3D shapes. The pipeline requires users to input a 3D spatial curve, but creating such curves accurately and efficiently can be challenging for users. To address this issue and unleash creativity, a parametric tool is designed to offer an intuitive way to generate space curves for initial reference, enabling beginners readily obtain valid and satisfactory curves. The tool adopts meshes available on the internet as input, and the output is contour curves that users can adjust to meet their needs and later feed into the pipeline to generate zippers. Based

on the output curves, the demo shows a design suite of curvilinear animal sculptures. We believe the demo imbues abstract curves with perceptible vitality that inspire artists to create even more beautiful curvilinear artworks.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; • **Computing methodologies** → **Shape modeling**; • **Applied computing** → **Computer-aided design**.

Keywords

Line-based structure, 3D printing, Decoration design, Personal fabrication, Rapid prototyping, Computational fabrication

*Corresponding author.

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1 Introduction

Linear structures are essential in engineering, art, and design. Electrical signals decompose into sinusoidal waves, while Picasso's *Le Taureau* uses curves to distill bulls into expressive, powerful forms [7]. Similarly, Calatrava's *The Oculus* embodies kinetic fluidity, evoking a bird in flight [1]. In the field of personal fabrication, the linear structure is also a popular research subject. The recent emergence of innovative technologies in digital fabrication has effectively bridged the gap between the design target and the fabrication result while lowering the threshold for the participation of the general audience. Computational design methods have accomplished various scales [3, 4, 6, 8] and configurations [6, 9–11, 13], some supporting inverse design to achieve target forms [3, 4, 11, 14]. Particularly, existing work Zipperables [9], RichRach [2] have provided novel assembly methods with zippers and zipper-like structures, and StructCurves [12] has also proposed 3D-printed joints for modular assembly.

Zipper structures offer an effective solution for materializing abstract curves or skeletal frameworks, combining linear and dimensional properties with detachable functionality. Their interactive nature allows flexible material edges to be joined and separated, while computational design methods enable 3D-printed zippers that maintain predefined shapes [5]. Touch-n-Curl presents a design and construction system based on FDM 3D printing that produces 3D skeletal forms through zipper-based 2D assemblies.

In this demo, we present animal sculptures in curvilinear style fabricated with 3D-printed zippers. Due to the steep learning curve and complexity involved in designing curves or skeletal structures in Rhino, our system primarily focuses on generating valid curves or curve trees for the Touch-n-Curl method. As shown in Figure 1, users can input a mesh and adjust cutting surfaces, after which our parametric tool automatically generates suitable curves or curve trees for subsequent Touch-n-Curl processing steps - including ribbon generation, teeth adjustment, and feature implement. The final zipper assemblies are manually interlocked into 3D configurations, resulting in expressive curvilinear animal sculptures.

2 Zipper Design

The zipper system combines profiles and ribbons, using tapered geometries (e.g., polygonal/conical frustums) for fastening. When arranged in alternating linear arrays, these profiles interlock to form strong connections.

For successful shape restoration, the design must adapt to various curvatures and torsions while precisely controlling profile engagement. Consequently, we 3D-print the profiles and ribbon as a single unit, minimizing overhangs that require supports.

To apply the same zipper design to create a space curve, we have conceptualized three types of modifications to the basic zipper design. We call them *elemental rotations* (Figure 2): (1) Rotation A refers to the curvature of the ribbon edges within the ribbon plane, which can go either left or right; (2) Rotation B enables the torsional motion around the tangent, going either clockwise or counter-clockwise; (3) Rotation C stands for the curvature of the ribbon moving away from the ribbon plane, which either goes

up or down. Three rotations are programmed into one piece of grasshopper algorithm.

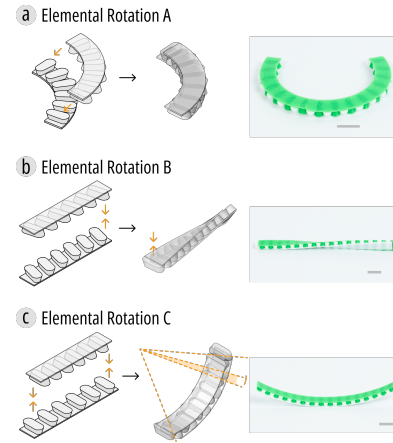


Figure 2: The three types of elemental rotations to produce a curved or twisted zipper assembly from a straight zipper. (Scale bar: 1 cm.)

3 Curve Generating

There are many methods to create curves, including parametric equations, point interpolation, sketches, etc. Using these tools, experienced users can manually design space curves to meet the requirements of the Touch-n-Curl generation tool. For novice users, we provide a Grasshopper-based design tool to automatically generate reference curves.

Our design tool takes a mesh as input (Figure 1a) and outputs a set of curves or a curve tree. When users input the initial mesh, the tool first converts the mesh to a SubD object and generates two orthogonal cutting planes by default. This produces two sets of intersection curves between the planes and the SubD object. Users can also create custom surfaces to split the mesh, obtaining more specific and featured intersection curves.

After generating the intersection curves, our design tool discretizes the curves at a given interval (10 mm by default), producing a point cloud. Based on the normal vectors of the original surface, each point extrudes a line penetrating the entire mesh. These lines are then partitioned into two curve sets according to the input surface. A new point cloud for the final skeleton is generated by computing the closest points between these two sets of lines.

Subsequently, the design tool generates a set of curves or a curve tree that represents the shape of the input mesh (Figure 1b), derived from these two sets of intersection curves. After baking these curves or curve tree from Grasshopper into Rhino, users can also adjust the curves to meet their own requirement.

4 Results and Discussion

Diversified animal sculptures are designed (Figure 1d). We have also realized that shadows of animal sculptures casting on the wall are also interesting. When people move around the light source, the shadows undergo changes as if animals were alive. This may be a potential design space for zipper structures.

During the prototyping process, it is worth noting that in the process of curve generating, user customization specifically the choice of cutting surfaces significantly impacts the resulting point cloud and curve morphology. The final shape fidelity depends not only on the geometry and placement of these cuts but also on the point cloud density (i.e., sampling interval).

5 Conclusion

This demo presents Touch-n-Curl, an integrated pipeline for creating curvilinear sculptures through computationally designed, 3D-printed zipper assemblies. By developing parametric tools that automatically generate space curves from mesh inputs, we lower the barrier for novice users while preserving the freedom for experts. Our zipper design—featuring tapered interlocking profiles and three elemental rotations—enables the transformation of 2D printed flat assemblies into complex 3D forms. The resulting animal sculptures not only validate the system’s technical feasibility but also reveal aesthetic potentials. This work contributes to the intersection of computational design and tangible fabrication, providing artists and designers with a novel approach to curvilinear expression. It also opens up possibilities for further exploration of interactive zipper structures in fields such as kinetic art and spatial design.

Acknowledgments

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